

Question Paper: Data Structures

Maximum Marks: 40

Q1. What is an Array? (Max Marks: 10)

Ans:

An array is a linear data structure used to store multiple elements of the same data type in contiguous memory locations. Each element can be accessed directly using an index. Arrays allow efficient access but have a fixed size, which limits flexibility.

Q2. Explain Stack and its operations. (Max Marks: 10)

Ans:

A stack is a linear data structure that follows the LIFO (Last In First Out) principle.

The main operations of stack are push, pop, and peek.

Push is used to insert an element, pop removes the top element,

and peek returns the top element without removing it.

Q3. What is a Linked List? (Max Marks: 10)

Ans:

A linked list is a dynamic data structure in which elements are stored in nodes.

Each node contains data and a reference to the next node.

Unlike arrays, linked lists do not store data in contiguous memory locations.

They allow efficient insertion and deletion of elements.

Q4. Explain Queue and its applications. (Max Marks: 10)

Ans:

A queue is a linear data structure that follows FIFO (First In First Out) principle.

Elements are inserted at the rear and removed from the front.

Queues are widely used in CPU scheduling, printer management,

and breadth-first search algorithms.

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